

# SEC P vs NP — Frameworks (live)

SEC (autonomous) — attributed to Ludovico Kubler

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Compiled 2026-04-27 17:00 UTC. Tracking 3 active + 0 dead frameworks.

### Summary table

| Framework ID  | Status      | Fitness | Generation | Parent | Name                                        |
|---------------|-------------|---------|------------|--------|---------------------------------------------|
| fw_b9e7d103d0 | ELABORATING | 0.000   | 0          | -      | Ergodic Circuit Framework                   |
| fw_28b4bfb95f | ELABORATING | 0.000   | 0          | -      | TROPICAL_FOURIER                            |
| fw_85a254b4a0 | ELABORATING | 0.000   | 0          | -      | Coarse Geometric Karchmer-Wigderson (CG-KW) |

### Details

#### Ergodic Circuit Framework (fw\_b9e7d103d0)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** COMM\_COMPLEXITY
- **Target invariant:** communication\_entropy\_barrier (real number  $\geq 0$ )  $\rightarrow$  bounds The minimum communication complexity in a  $k$ -party number-on-forehead model, where lower bounds are derived from the growth rate of Kolmogorov-Sinai entropy in associated dynamical circuits. Aims to prove  $\omega(\log n)$  lower bounds for explicit functions like Disjointness, avoiding natural proofs via non-uniform dynamics.

**Primitives:** - measurable\_dynamical\_circuit (tuple  $(C, X, \mu, T)$ ): A Boolean circuit  $C$  where gates are labeled by operations over a probability space  $(X, \mu)$ , and  $T: X \rightarrow X$  is a measure-preserving transformation  
- orbit\_signature (function:  $N \rightarrow [0,1]$ ): For a fixed input  $x \in \{0,1\}^n$ , the orbit signature is the sequence  $(\mu(C(T^k(x))))_k$ , capturing how the circuit's output evolves under repeated application  
- cross\_correlation\_flow (matrix  $\in R^{d \times d}$ ): For a depth- $d$  circuit, this matrix  $F\{i,j\} = \int |C_i(x) - C_i(T^{j(x)}(x))|^2 d\mu(x)$  measures how perturbations via  $T$  propagate through layers.  
Computable via

**Tentative axioms:** - A1: Any circuit with  $o(n)$  communication complexity induces a dynamical system with sublinear Kolmogorov-Sinai entropy growth. - A2: Measure-preserving transformations corresponding to  $AC^0$  circuits have bounded mixing time, while those for PSPACE-complete computations exhibit super-polynomial mixer-profile decay. - A3: If a family of circuits computes a function with high cross\_correlation\_flow norm, then its communication complexity is  $\Omega(n^\delta)$  for some  $\delta > 0$ .

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## TROPICAL\_FOURIER\_ANALYSIS (fw\_28b4bfb95f)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** FOURIER\_ANALYTIC
- **Target invariant:** MinimalFourierCoefficient (RealNumber)  $\rightarrow$  bounds The discrepancy of a function under tropical Fourier analysis.

**Primitives:** - TropicalPolynomial (Function): A function defined over a tropical semiring (max-plus or min-plus) with computable coefficients. - TropicalFourierTransform (Transformation): A mapping from tropical polynomials to their Fourier coefficients over a tropical semiring. - DiscrepancyMeasure (Metric): A function quantifying the deviation of a tropical polynomial from uniformity.

**Tentative axioms:** - A1: TropicalConvolution preserves the tropical semiring structure under composition. - A2: TropicalFourierTransform is invertible when restricted to certain tropical polynomials. - A3: DiscrepancyMeasure is bounded by the maximum absolute value of Fourier coefficients.

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## Coarse Geometric Karchmer-Wigderson (CG-KW) (fw\_85a254b4a0)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** KARCHMER\_WIGDERSON
- **Target invariant:** Coarse depth  $\kappa(f)$  (non-negative real)  $\rightarrow$  bounds  $\kappa(f) := \sup$  over nontrivial  $c \in HX^1(X_f)$  and  $T \in C_u^*[X_f]$  of  $\log|\langle c, T \rangle| / \log(\text{propagation}(T))$ . The conjecture is  $\text{depth}(f) \geq \kappa(f)$ , so a super-logarithmic  $\kappa(f)$  for an explicit  $f$  gives a super-logarithmic formula-depth lower bound (a  $P \not\subseteq NC^1$  separation if  $\kappa$  is poly-large).

**Primitives:** - KW-metric space (finite metric space  $(X_f, d_f)$  with  $X_f = f^{-1}\{0\} \sqcup f^{-1}\{1\}$ ): For a boolean function  $f: \{0,1\}^n \rightarrow \{0,1\}$ , define  $d_f(x,y) = \log_2(\text{min number of coordinates a depth-bounded distinguisher must read to certify } f(x) \neq f(y))$ . - Controlled cover (finite family  $U = \{U_i \subset X_f\}$  with diameter bound  $R$ ): A cover whose parts have  $d_f$ -diameter at most  $R$  and whose overlap multiplicity is at most  $m$ . Encoded as a hypergraph  $H_U$  (vertices =  $X_f$ , edges =  $U_i$ ). - Coarse cocycle (function  $c: X_f \times X_f \rightarrow \mathbb{Z}$  with finite support modulo controlled equivalence): A discrete 1-cochain in the Roe-style coarse cohomology  $HX^1(X_f)$  restricted to the KW pair structure. Stored as a sparse table on pairs  $(x,y)$  with  $d_f(x,y) \leq R$ . - Asymptotic-dimension witness (vertex coloring  $\chi: X_f \rightarrow [k]$  together with diameter bound  $R$ ): Witnesses  $\text{asdim}(X_f) \leq k-1$  in the sense of Gromov: each color class decomposes into  $d_f$ -bounded pieces of diameter  $\leq R$  that are pairwise  $R$ -disjoint. - Roe-controlled operator (matrix  $T \in R^{\{X_f \times X_f\}}$  with  $T_{\{x,y\}} = 0$  whenever  $d_f(x,y) > R$ ): Element of the algebraic uniform Roe algebra  $C_u^*[X_f]$ . Stored as a banded sparse matrix in the  $d_f$ -metric; multiplication, adjoint, and trace are al

**Tentative axioms:** - A1 (Coarse-KW link): formula depth  $d(f) \geq \Omega(\log(\text{asdim}(X_f)))$  and more strongly  $d(f) \geq \Omega(\kappa(f))$ ; proved by translating a KW protocol into a controlled cover whose multiplicity exponent equals depth. - A2 (Composition sub-additivity):  $\text{asdim}(X_{\{f \circ g\}}) \geq \text{asdim}(X_f) + \text{asdim}(X_g) - O(1)$ , giving a KRW-style direct-sum theorem at the level of coarse dimension. - A3 (Anti-natural-proofs): for a uniformly random  $f$ ,  $HX^1(X_f)$  is generically zero (Roe-algebraic concentration of measure), so  $\kappa$  is NOT a 'largeness' property in the Razborov-Rudich sense — random  $f$ 's - A4 (Anti-relativization):  $\kappa$  is a metric invariant of  $d_f$  and is destroyed by oracle

access (oracle gates collapse  $d_f$  to  $\leq 1$ ), so  $\kappa$ -based bounds cannot relativize, in the spirit of Mendel-Naor metric - A5 (Roe-index rigidity): nontrivial classes in  $HX^1(X_f)$  detected by the Roe-trace pairing are stable under coarse equivalence, mirroring Yu's coarse Baum-Connes results (Yu 2000; Nowak-Yu 2012); expl